

Calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

You have the entire period.

Signature

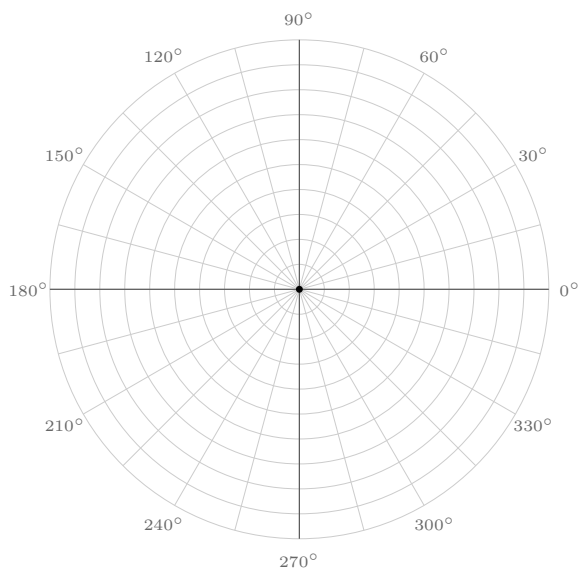
SHOW ALL WORK FOR FULL CREDIT!

Show your set up (what you are putting into the calculator).

Use your calculator to sketch a graph of each polar equation. Identify the type of polar graph it is. (3 points each)

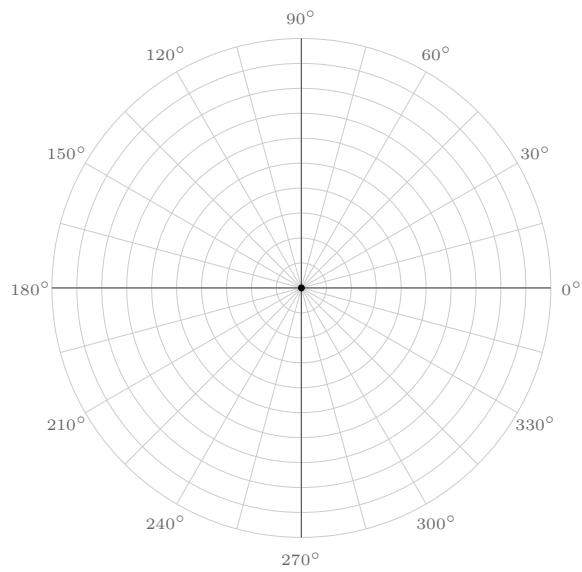
Word bank: circle cardioid limaçon rose curve lemniscate

1. $r = 5 - 5 \cos \theta$



Type: _____

2. $r = 4 \sin 3\theta$



Type: _____

*Non-Calculator***SHOW ALL WORK FOR FULL CREDIT!**

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

3.

a) -8

b) $3 + \sqrt{-16}$

Find the product. Write your answer in standard form. (2 points)

4. $\sqrt{-8} \cdot \sqrt{-50}$

Simplify each power of i . (1 point each)

5.

a) i^{53}

b) $\frac{1}{i^{45}}$

Find the product. Write your answer in standard form. (2 points)

6. $(2 + 7i)(3 - i)$

Find the quotient. Write your answer in standard form. (3 points)

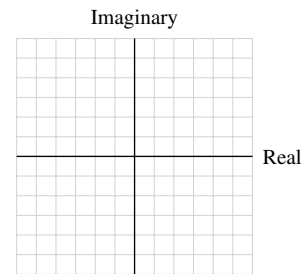
7. $\frac{7 + i}{1 + i}$

Solve the equation over the set of complex numbers. (3 points)

8. $3x^2 + 2 = 2x$

Find the sum of the pair of complex numbers. Write your answer in standard form. Then graph both complex numbers and their resultant, and find the absolute value of the sum. (2 points)

9. $-3 + 4i$, $5 - 2i$



Write the complex number in rectangular form. (2 points)

10. $4(\cos 225^\circ + i \sin 225^\circ)$

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

11.

a) $-3\sqrt{3} + 3i$

b) -5

The Julia set: compute $z^2 - 1$, $(z^2 - 1)^2 - 1$, $[(z^2 - 1)^2 - 1]^2 - 1, \dots$. If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. Determine whether each number belongs to the Julia set. (1 point each)

12.

a) $z = -1$

YES / NO

b) $z = 2i$

YES / NO

Find the product. Write the product in rectangular form, using exact values. (3 points)

13. $[\sqrt{6} (\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12})] [\sqrt{6} (\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

14. $\frac{\sqrt{6} - i\sqrt{2}}{-1 - i\sqrt{3}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

15. $(-\sqrt{3} + i)^3$

Find all n th roots. Write your answers in polar form. (3 points)

16. $-8 + 8i\sqrt{3}$, $n = 4$

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

17.

a) $z = -2$

YES / NO

b) $z = 1 - i$

YES / NO

18. For the point $(-8, \frac{3\pi}{4})$ in polar coordinates, (a) state the quadrant in which the point lies if it is graphed in a rectangular coordinate system, (b) determine which is **not** an equivalent point, and (c) find the rectangular coordinates of the point in exact form. 3 pts

a) $(8, \frac{7\pi}{4})$

b) $(-8, -\frac{5\pi}{4})$

c) $(8, -\frac{\pi}{4})$

d) $(8, \frac{3\pi}{4})$

19. For the rectangular coordinates $(3, -3\sqrt{3})$, give two pairs of polar coordinates for the point, where $0^\circ \leq \theta < 360^\circ$. 3 pts

Convert each rectangular equation to a polar equation. (2 points each)

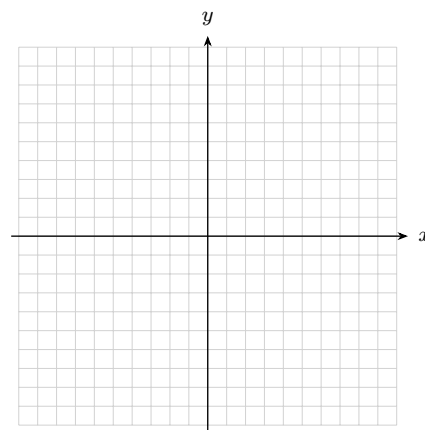
20.

a) $5x + 2y = 8$

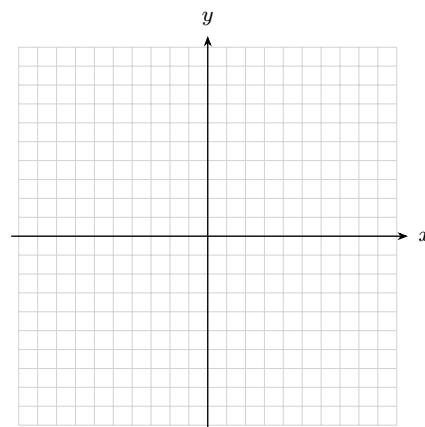
b) $x^2 + y^2 = 12$

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

21. $r = 8 \cos \theta - 6 \sin \theta$



22. $r = \frac{6}{1 - \cos \theta}$



23. Find the polar equation of the line that passes through $(-3, 0^\circ)$ and $(6, 90^\circ)$. Simplify your answer.

3 pts

Find the polar coordinates of the points of intersection of the given curves for the specified interval of θ . (4 points)

24. $r = 3 + 2 \cos \theta$, $r = 3 - 2 \sin \theta$, $0 \leq \theta < 2\pi$

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (1 point each)

25.

a) $r = 3 \cos 4\theta$ _____

b) $r = 6 - 3 \sin \theta$ _____

c) $r = 5 - 5 \sin \theta$ _____

d) $r^2 = 16 \sin 2\theta$ _____

e) $r = 3 \sin \theta$ _____

ANSWER KEY – Calculator**SHOW ALL WORK FOR FULL CREDIT!**

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

1. $r = 5 - 5 \cos \theta$

WORK

Polar mode: $r1(\theta) = 5 - 5 \cos \theta$

$\theta = 0^\circ$: $r = 5 - 5 = 0$

$\theta = 90^\circ, 270^\circ$: $r = 5$

$\theta = 180^\circ$: $r = 5 - 5(-1) = 10$

$\frac{a}{b} = \frac{5}{5} = 1$

cardioid

WHY

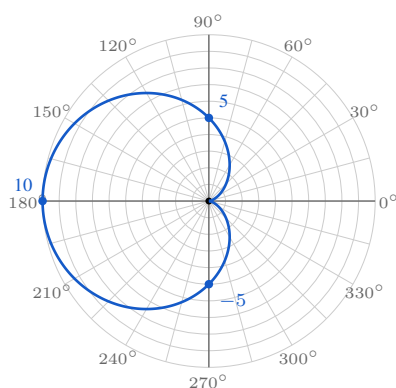
Check the angle mode if nothing shows.

Touches the pole: the dent of the heart.

Crosses the y -axis at 5 and -5 .

Farthest point, $(-10, 0)$.

Ratio exactly 1.



2. $r = 4 \sin 3\theta$

WORK

Polar mode: $r1(\theta) = 4 \sin(3\theta)$

$n = 3$ (odd): 3 petals

$\theta = 30^\circ$: $r = 4 \sin 90^\circ = 4$

tips at $30^\circ, 150^\circ, 270^\circ$

rose curve

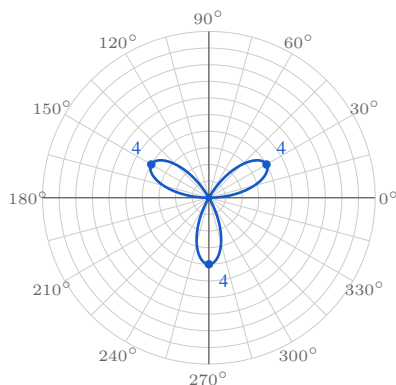
WHY

Type the parentheses around 3θ .

n odd: n petals. n even: $2n$.

Petal length = $|A| = 4$.

At 90° , $r = -4$ plots at 270° .



ANSWER KEY – Non-Calculator**SHOW ALL WORK FOR FULL CREDIT!**

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Identify each number as *real*, *complex*, *pure imaginary*, or *nonreal complex*. (More than one of these descriptions may apply.) (2 points)

3.

WORK

a) $-8 = -8 + 0i$

real, complex

b) $3 + \sqrt{-16} = 3 + 4i$

complex, nonreal complex

WHY

$b = 0.$

Real numbers are complex too.

$a \neq 0, b \neq 0.$

Not pure imaginary: $a \neq 0.$

Find the product. Write your answer in standard form. (2 points)

4. $\sqrt{-8} \cdot \sqrt{-50}$

WORK

$\sqrt{-8} = 2i\sqrt{2}, \sqrt{-50} = 5i\sqrt{2}$

$(2i\sqrt{2})(5i\sqrt{2}) = 10i^2(2)$

$= 20i^2 = -20$

Trap: $\sqrt{(-8)(-50)} = \sqrt{400} = 20$ has the wrong sign.

WHY

Pull out i first.

Multiply the pieces.

$i^2 = -1.$

Simplify each power of i . (1 point each)

5.

WORK

a) $53 \div 4 = 13 \text{ R } 1, i^{53} = i^1 = i$

b) $45 \div 4 = 11 \text{ R } 1, i^{45} = i$

$\frac{1}{i} \cdot \frac{i}{i} = \frac{i}{i^2} = \frac{i}{-1} = -i$

WHY

Only the remainder matters.

Clear i from the bottom.

Find the product. Write your answer in standard form. (2 points)

6. $(2 + 7i)(3 - i)$

WORK

$6 - 2i + 21i - 7i^2$

$= 6 + 19i + 7$

$= 13 + 19i$

WHY

FOIL.

$-7i^2 = +7.$

Find the quotient. Write your answer in standard form. (3 points)

7. $\frac{7+i}{1+i}$

WORK

$$\frac{7+i}{1+i} \cdot \frac{1-i}{1-i}$$

top: $7 - 7i + i - i^2 = 8 - 6i$

bottom: $1 + 1 = 2$

$$\frac{8-6i}{2} = \boxed{4-3i}$$

WHY

Multiply by the conjugate of the bottom.

$$-i^2 = +1.$$

$$a^2 + b^2.$$

Solve the equation over the set of complex numbers. (3 points)

8. $3x^2 + 2 = 2x$

WORK

$$3x^2 - 2x + 2 = 0$$

$$x = \frac{2 \pm \sqrt{4 - 4(3)(2)}}{2(3)} = \frac{2 \pm \sqrt{-20}}{6}$$

$$= \frac{2 \pm 2i\sqrt{5}}{6}$$

$$\boxed{x = \frac{1}{3} \pm \frac{\sqrt{5}}{3}i}$$

WHY

Zero on one side first.

$$a = 3, b = -2, c = 2; -b = +2.$$

$$\sqrt{-20} = 2i\sqrt{5}.$$

Reduce both parts.

Find the sum of the pair of complex numbers. Write your answer in standard form. Then graph both complex numbers and their resultant, and find the absolute value of the sum. (2 points)

9. $-3 + 4i, 5 - 2i$

WORK

$$(-3 + 5) + (4 - 2)i = \boxed{2 + 2i}$$

$$|2 + 2i| = \sqrt{4 + 4} = \sqrt{8} = \boxed{2\sqrt{2}}$$

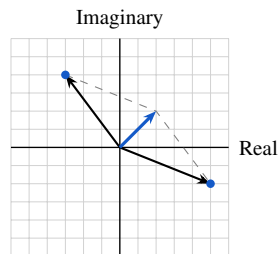
Graph: $(-3, 4), (5, -2)$, resultant $(2, 2)$

WHY

Add reals, add imaginaries.

Simplify the radical.

Diagonal of the parallelogram.



Write the complex number in rectangular form. (2 points)

10. $4(\cos 225^\circ + i \sin 225^\circ)$

WORK

$$4\left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right)$$

$$= \boxed{-2\sqrt{2} - 2\sqrt{2}i}$$

WHY

225° is QIII: both negative.

Write each complex number in trigonometric (polar) form, with θ in the interval $[0^\circ, 360^\circ)$. (2 points each)

11.

WORK

$$a) r = \sqrt{27 + 9} = 6$$

$$\theta = \tan^{-1}\left(\frac{3}{-3\sqrt{3}}\right) = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -30^\circ$$

$$\theta = -30^\circ + 180^\circ = 150^\circ$$

$$\boxed{6 \operatorname{cis} 150^\circ}$$

$$b) -5 = -5 + 0i: r = 5, \theta = 180^\circ$$

$$\boxed{5 \operatorname{cis} 180^\circ}$$

WHY

$$r = \sqrt{x^2 + y^2}.$$

$\tan^{-1}$ only gives QI or QIV.

$x < 0, y > 0$ is QII: add 180° .

On the negative real axis.

The Julia set: compute $z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$. If the absolute value of any result exceeds 2, z is **not** in the Julia set; otherwise it is. Determine whether each number belongs to the Julia set. (1 point each)

12.

WORK

$$a) (-1)^2 - 1 = 0, 0^2 - 1 = -1$$

$\boxed{\text{YES}}$

$$b) (2i)^2 - 1 = -4 - 1 = -5$$

$$|-5| = 5 > 2: \boxed{\text{NO}}$$

WHY

Cycles $0, -1, 0, \dots$; never over 2.

$$(2i)^2 = 4i^2 = -4.$$

Find the product. Write the product in rectangular form, using exact values. (3 points)

$$13. \left[\sqrt{6} \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right] \left[\sqrt{6} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right) \right]$$

WORK

$$\sqrt{6} \cdot \sqrt{6} \operatorname{cis} \left(\frac{7\pi}{12} + \frac{9\pi}{12} \right)$$

$$= 6 \operatorname{cis} \frac{16\pi}{12} = 6 \operatorname{cis} \frac{4\pi}{3}$$

$$= 6 \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right)$$

$$= \boxed{-3 - 3\sqrt{3}i}$$

WHY

Product theorem. Common denominator:

$$\frac{3\pi}{4} = \frac{9\pi}{12}.$$

Reduce the fraction.

$$\frac{4\pi}{3} \text{ is QIII.}$$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

$$14. \frac{\sqrt{6} - i\sqrt{2}}{-1 - i\sqrt{3}}$$

WORK

$$r_1 = \sqrt{6 + 2} = 2\sqrt{2}$$

$$\theta_1 = \tan^{-1}\left(\frac{-\sqrt{2}}{\sqrt{6}}\right) = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -30^\circ \rightarrow 330^\circ$$

$$r_2 = \sqrt{1 + 3} = 2, \theta_2 = \tan^{-1}(\sqrt{3}) = 60^\circ$$

$$\theta_2 = 60^\circ + 180^\circ = 240^\circ$$

$$\frac{2\sqrt{2} \operatorname{cis} 330^\circ}{2 \operatorname{cis} 240^\circ} = \sqrt{2} \operatorname{cis} 90^\circ$$

$$= \sqrt{2}(0 + 1i) = \boxed{\sqrt{2}i}$$

WHY

Top first.

$$\frac{\sqrt{2}}{\sqrt{6}} = \frac{1}{\sqrt{3}}. \text{ QIV: add } 360^\circ.$$

Reference angle.

$x < 0, y < 0$ is QIII.

Quotient theorem.

$$\cos 90^\circ = 0, \sin 90^\circ = 1.$$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

15. $(-\sqrt{3} + i)^3$

WORK

$$r = \sqrt{3 + 1} = 2$$

$$\theta = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -30^\circ, -30^\circ + 180^\circ = 150^\circ$$

$$(2 \operatorname{cis} 150^\circ)^3 = 8 \operatorname{cis} 450^\circ$$

$$= 8 \operatorname{cis} 90^\circ = \boxed{8i}$$

WHY

$x < 0, y > 0$ is QII.

De Moivre.

$$450^\circ - 360^\circ = 90^\circ.$$

Find all n th roots. Write your answers in polar form. (3 points)

16. $-8 + 8i\sqrt{3}, \quad n = 4$

WORK

$$r = \sqrt{64 + 192} = \sqrt{256} = 16$$

$$\theta = \tan^{-1}(-\sqrt{3}) = -60^\circ, -60^\circ + 180^\circ = 120^\circ$$

$$\sqrt[4]{16} = 2$$

$$\alpha = \frac{120^\circ + 360^\circ k}{4} = 30^\circ + 90^\circ k$$

$$\boxed{2 \operatorname{cis} 30^\circ, 2 \operatorname{cis} 120^\circ, 2 \operatorname{cis} 210^\circ, 2 \operatorname{cis} 300^\circ}$$

WHY

$$(8\sqrt{3})^2 = 64 \cdot 3 = 192.$$

$x < 0, y > 0$ is QII.

n th root theorem: root of r .

$k = 0, 1, 2, 3$: four roots, 90° apart.

Polar form was asked, so stop here.

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

17.

WORK

$$\text{a) } -2, (-2)^2 + (-2) = 2, 2^2 + (-2) = 2$$

$\boxed{\text{YES}}$

$$\text{b) } (1 - i)^2 + (1 - i) = -2i + 1 - i = 1 - 3i$$

$$|1 - 3i| = \sqrt{1 + 9} = \sqrt{10} \approx 3.16 > 2: \boxed{\text{NO}}$$

WHY

Stuck at 2 forever.

$$|2| = 2 \text{ does not exceed } 2.$$

$$(1 - i)^2 = 1 - 2i + i^2 = -2i.$$

18. For the point $(-8, \frac{3\pi}{4})$ in polar coordinates, (a) state the quadrant in which the point lies if it is graphed in a rectangular coordinate system, (b) determine which is **not** an equivalent point, and (c) find the rectangular coordinates of the point in exact form.

WORK

$$\text{(a) face } \frac{3\pi}{4} \text{ (QII), walk backward: } \boxed{\text{QIV}}$$

$$\text{(b) } (-8, \frac{3\pi}{4}) = (8, \frac{3\pi}{4} + \pi) = (8, \frac{7\pi}{4})$$

$$(8, -\frac{\pi}{4}) \text{ same as } \frac{7\pi}{4}; -\frac{5\pi}{4} \text{ same as } \frac{3\pi}{4}$$

$$(8, \frac{3\pi}{4}) \text{ is in QII: } \boxed{\text{d}}$$

$$\text{(c) } x = -8 \cos \frac{3\pi}{4} = -8 \left(-\frac{\sqrt{2}}{2}\right) = 4\sqrt{2}$$

$$y = -8 \sin \frac{3\pi}{4} = -8 \left(\frac{\sqrt{2}}{2}\right) = -4\sqrt{2}$$

$$\boxed{(4\sqrt{2}, -4\sqrt{2})}$$

WHY

Negative r lands in the opposite quadrant.

$$(-r, \theta) = (r, \theta \pm \pi).$$

$\pm 2\pi$ keeps the point.

Keep r 's sign.

QIV, matches (a).

19. For the rectangular coordinates $(3, -3\sqrt{3})$, give two pairs of polar coordinates for the point, where $0^\circ \leq \theta < 360^\circ$.

WORK

$$r = \sqrt{9 + 27} = 6$$

$$\tan \theta = \frac{-3\sqrt{3}}{3} = -\sqrt{3}, \theta = -60^\circ$$

$$\theta = -60^\circ + 360^\circ = 300^\circ$$

$$(6, 300^\circ), (-6, 120^\circ)$$

WHY

$$r^2 = x^2 + y^2.$$

$\tan^{-1}$ answer.

$x > 0, y < 0$: QIV.

Second pair: $(-r, \theta - 180^\circ)$.

Convert each rectangular equation to a polar equation. (2 points each)

20.

WORK

a) $5r \cos \theta + 2r \sin \theta = 8$

$$r = \frac{8}{5 \cos \theta + 2 \sin \theta}$$

b) $r^2(\cos^2 \theta + \sin^2 \theta) = 12, r^2 = 12$

$$r = \pm\sqrt{12} = \pm 2\sqrt{3}$$

$$r = 2\sqrt{3} \text{ or } r = -2\sqrt{3}$$

WHY

Substitute.

Factor r , divide.

Pythagorean identity.

$\pm$, then simplify.

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

21. $r = 8 \cos \theta - 6 \sin \theta$

WORK

$$r^2 = 8r \cos \theta - 6r \sin \theta$$

$$x^2 + y^2 = 8x - 6y$$

$$x^2 - 8x + 16 + y^2 + 6y + 9 = 16 + 9$$

$$(x - 4)^2 + (y + 3)^2 = 25$$

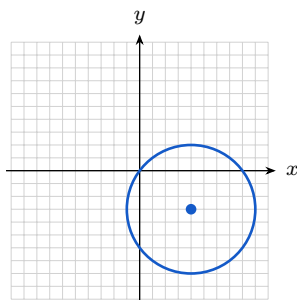
WHY

Multiply both sides by r .

Substitute.

Complete the square twice.

Circle, center $(4, -3)$, radius 5.



22. $r = \frac{6}{1 - \cos \theta}$

WORK

$$r(1 - \cos \theta) = 6, \quad r - r \cos \theta = 6$$

$$r - x = 6, \quad r = x + 6$$

$$x^2 + y^2 = x^2 + 12x + 36$$

$$y^2 = 12x + 36$$

$$y^2 = 12(x + 3)$$

WHY

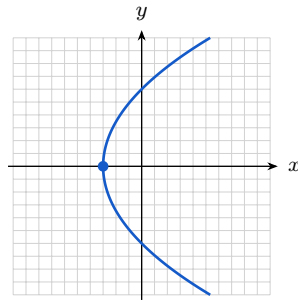
Clear the fraction.

$r \cos \theta = x$; isolate r .

Square both sides; $r^2 = x^2 + y^2$.

x^2 cancels.

Sideways parabola, vertex $(-3, 0)$, opens right.



23. Find the polar equation of the line that passes through $(-3, 0^\circ)$ and $(6, 90^\circ)$. Simplify your answer.

WORK

$$(-3, 0^\circ): x = -3 \cos 0^\circ = -3, y = 0, \quad (-3, 0)$$

$$(6, 90^\circ): x = 0, y = 6, \quad (0, 6)$$

$$m = \frac{6 - 0}{0 - (-3)} = 2, \quad y = 2x + 6$$

$$2x - y = -6$$

$$2r \cos \theta - r \sin \theta = -6$$

$$r = \frac{-6}{2 \cos \theta - \sin \theta}$$

WHY

Negative r puts it on the left.

Slope, y -intercept 6.

Standard form.

Substitute.

Factor r , divide.

Find the polar coordinates of the points of intersection of the given curves for the specified interval of θ . (4 points)

24. $r = 3 + 2 \cos \theta, \quad r = 3 - 2 \sin \theta, \quad 0 \leq \theta < 2\pi$

WORK

$$3 + 2 \cos \theta = 3 - 2 \sin \theta$$

$$\sin \theta = -\cos \theta$$

$$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$\theta = \frac{3\pi}{4}: r = 3 + 2 \left(-\frac{\sqrt{2}}{2} \right) = 3 - \sqrt{2}$$

$$\theta = \frac{7\pi}{4}: r = 3 + 2 \left(\frac{\sqrt{2}}{2} \right) = 3 + \sqrt{2}$$

$$\left(3 - \sqrt{2}, \frac{3\pi}{4} \right), \left(3 + \sqrt{2}, \frac{7\pi}{4} \right)$$

WHY

Set $r = r$.

Do not divide by $\cos \theta$.

Unit circle points where $y = -x$.

Plug back in.

Neither curve reaches the pole ($r \geq 1$).

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (1 point each)

25.

WORK

- a)
- b)
- c)
- d)
- e)

WHY

$n = 4$ even: $2n = 8$ petals, each 3 long.

$$\frac{a}{b} = \frac{6}{3} = 2 \neq 1.$$

$$\frac{a}{b} = \frac{5}{5} = 1.$$

r^2 equation.

$$r^2 = 3r \sin \theta \text{ gives } x^2 + y^2 = 3y.$$